

J MUKHESH

2303A51813

Frog jump

```
class Solution {  
    int minCost(int[] height) {  
        int n = height.length;  
        if(n == 1)  
            return 0;  
        int[] dp = new int[n];  
        dp[0] = 0;  
        dp[1] = Math.abs(height[0] - height[1]);  
        for(int i = 2; i < n; i++){  
            int i1 = Math.abs(height[i] - height[i - 1]);  
            int i2 = Math.abs(height[i] - height[i - 2]);  
            dp[i] = Math.min(i1 + dp[i - 1], i2 + dp[i - 2]);  
        }  
        return dp[n - 1];  
    }  
}
```

SQL

1. SELECT e.employee_id, e.employee_name, d.department_name, d.location
FROM employees e INNER JOIN departments d ON e.department_id = d.department_id;
2. SELECT e.employee_id, e.employee_name, d.department_name
FROM employees e LEFT JOIN departments d ON e.department_id = d.department_id;
3. SELECT d.department_id, d.department_name
FROM departments d LEFT JOIN employees e ON d.department_id = e.department_id
WHERE e.employee_id IS NULL;
4. SELECT e.employee_name, p.project_name, d.department_name, ep.hours_worked

FROM employees e JOIN departments d ON e.department_id=d.department_id JOIN employee_projects ep ON e.employee_id=ep.employee_id JOIN projects p ON ep.project_id=p.project_id;

5. SELECT p.project_id, p.project_name, e.employee_name

FROM projects p LEFT JOIN employee_projects ep ON p.project_id=ep.project_id LEFT JOIN employees e ON ep.employee_id=e.employee_id;

6. SELECT e.employee_name AS employee, m.employee_name AS manager

FROM employees e LEFT JOIN employees m ON e.manager_id=m.employee_id;

7. SELECT e.employee_id, e.employee_name, COUNT(ep.project_id) AS total_projects

FROM employees e JOIN employee_projects ep ON e.employee_id=ep.employee_id GROUP BY e.employee_id, e.employee_name HAVING COUNT(ep.project_id)>1;

8. SELECT d.department_name, COUNT(e.employee_id) AS employee_count, AVG(e.salary) AS average_salary FROM departments d LEFT JOIN employees e ON d.department_id=e.department_id GROUP BY d.department_id, d.department_name;